Learning to Diversify and Focus: A Reinforcement Framework for Open-Vocabulary HOI Detection

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Abstract

Open-Vocabulary Human–Object Interaction (OV-HOI) detection aims to recognize novel HOI categories beyond the training set. Existing OV-HOI detection approaches typically leverage CLIP to extract global visual representations and perform cross-attention between learnable queries and global features to localize human–object pairs. However, such one-stage paradigms tend to overfit seen interactions, limiting their generalization to unseen categories, while the coarse spatial awareness of CLIP also hinders the localization of fine-grained interaction cues. To address these issues, we propose a novel Semantic-Diversified and Interaction-Focused framework (SD-IF), which integrates reinforcement-guided adaptive optimization to jointly enhance semantic generalization and spatial discrimination. Specifically, we introduce a Semantic Diversification (SD) module that applies reinforcement-driven stochastic semantic perturbations and dual-level semantic exploration, expanding the semantic coverage of queries while maintaining visual coherence and effectively encouraging exploration beyond the seen semantic clusters. Furthermore, we design an Interaction Focusing (IF) module that formulates an actor–critic optimization scheme to adaptively refine attention distributions based on detection features and interaction representations, guided by a hybrid reward combining spatial focusing and semantic consistency. This cooperative learning paradigm enables the model to capture discriminative interaction cues and achieve spatially interpretable reasoning. Extensive experiments on two widely used benchmarks demonstrate that SD-IF achieves state-of-the-art performance, significantly surpassing existing OV-HOI detection methods.

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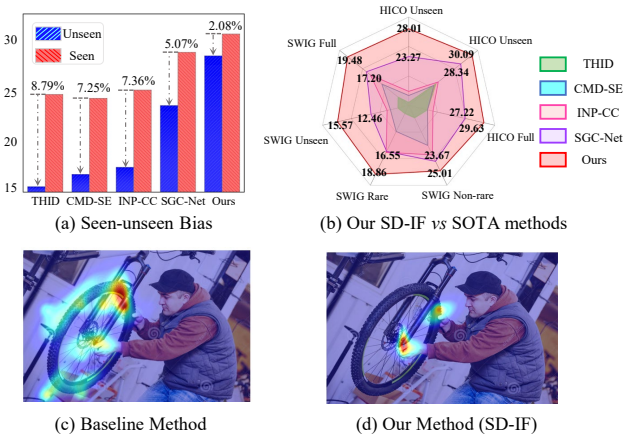


(a) Seen-unseen Bias (b) Our SD-IF *vs* SOTA methods

(c) Baseline Method (d) Our Method (SD-IF)

Figure 1. (a) Existing OV-HOI detection methods suffer from the limitations of the one-stage paradigm and tend to overfit seen HOIs , leading to substantial performance gaps between seen and unseen classes. (b) Our SD-IF framework consistently surpasses existing methods across all open-vocabulary metrics, demonstrating strong interaction understanding and generalization ability. (c) The baseline model's attention is scattered around the object. (d) Our SD-IF accurately focuses on the interaction-relevant regions.

1. Introduction

Human-Object Interaction (HOI) detection aims to localize the interacting human-object pairs and identify their interactions. It plays a crucial role in various computer vision applications, such as video analysis [40, 45], assistive robotics [26, 27], and scene understanding [5, 29]. Conventional HOI detection approaches can be broadly categorized into two paradigms based on the sequence of object detection and interaction classification sub-tasks: one-stage methods [21, 42] and two-stage methods [34, 44]. One-stage methods jointly perform object detection and interaction classification to directly generate HOI triplets, whereas two-stage methods explicitly decouple these sub-tasks, using detection results as positional priors for inter-

action recognition. However, both paradigms rely heavily on pre-trained object detectors and fixed category vocabularies, which limit their ability to generalize to novel HOI concepts in real-world scenarios [17, 37].

Recently, fueled by the rich world knowledge embedded in Vision–Language Models (VLMs) [20, 30], Open-Vocabulary Human–Object Interaction (OV-HOI) detection has achieved remarkable progress in recognizing unseen categories within open-world scenes [17, 18, 23, 37]. Existing OV-HOI approaches typically follow the one-stage paradigm, employing CLIP [30] as the image encoder to extract global visual representations and utilizing learnable queries within a transformer decoder to localize human–object pairs. Despite these advances, two major challenges remain. **(1) Seen-unseen Bias**: Due to the bipartite matching mechanism [3], the learnable queries are optimized primarily toward seen HOIs during training, leading to severe overfitting to the seen categories [16, 25] and a notable performance gap between seen and unseen classes. As shown in Figure 1 (a), the mAP gap exceeds 5%, highlighting the limited generalization to unseen categories. **(2) Limited Interaction Awareness**: In the open-vocabulary setting, the absence of a pre-trained object detector prevents explicit region proposals for human–object pairs. Moreover, the CLIP visual encoder, trained mainly on image-level supervision [6, 19], focuses on global semantics while neglecting region-level spatial cues essential for recognizing interactions. As shown in Figure 1 (c), the model's attention tends to disperse over irrelevant regions instead of focusing on the interaction-critical areas, resulting in sub-optimal feature alignment with textual semantics and degraded recognition accuracy.

To address these issues, we propose a novel Semantic-Diversified and Interaction-Focused framework (SD-IF), which employs reinforcement-guided adaptive optimization to jointly enhance **semantic generalization** and **spatial focusing**. As shown in Figure 1, our method alleviates the seen-unseen bias (Figure 1 (a)) and significantly outperforms existing SOTA methods (Figure 1 (b)) in various open vocabulary metrics.

First, to tackle the seen–unseen bias, we propose a Semantic Diversification (SD) module that enhances the semantic expressiveness of query embeddings within the decoder through reinforcement-driven stochastic perturbations. During training, each learnable query acts as an agent that samples semantic perturbations from a conditional Gaussian policy. The perturbed queries explore novel semantic directions under the guidance of the global visual context, allowing them to diversify while remaining visually coherent. To further promote open-set generalization, a proxy semantic reward is designed to encourage the perturbed queries to deviate from the centers of seen semantic clusters while preserving alignment with visual semantics. Through this **dual-level semantic exploration**, combining locally visual-consistent exploration with globally semantic expansion, SD effectively drives the query manifold toward unexplored regions of the semantic space, enabling the model to generalize to unseen HOI categories.

Second, to enhance spatial discrimination and strengthen interaction grounding, we propose an Interaction Focusing (IF) module formulated under an actor–critic reinforcement framework. Specifically, IF establishes an adaptive bridge between detection features and interaction semantics by learning an attention policy that highlights interaction-critical regions. During training, the actor network dynamically predicts spatial attention conditioned on perturbed queries, detection predictions, and interaction representations, while the critic network evaluates these attention maps through a **hybrid reward** that jointly measures spatial focusing and semantic consistency. As shown in Figure 1(d), this cooperative optimization encourages the model to concentrate attention within interaction regions and suppress background noise, thereby yielding spatially interpretable and discriminative HOI representations.

Our contributions are summarized below. (1) We introduce a Semantic Diversification Module that employs reinforcement-driven stochastic perturbations to expand the semantic coverage of query embeddings, significantly enhancing open-set generalization; (2) We design an Interaction Focusing Module that integrates detection and interaction cues through an actor–critic reinforcement learning scheme to dynamically focus on interaction-critical regions. (3) Our method outperforms previous state-of-the-art methods by a large margin on two public benchmarks HICO-DET [4] and SWIG-HOI [36], improving the network's adaptability to various novel interactions.

2. Related Work

2.1. Conventional HOI detection

Conventional HOI detection approaches can be broadly categorized into two pipelines: one-stage paradigm and two-stage paradigm. Early CNN-based one-stage methods associate humans and objects through auxiliary information such as interaction points [21, 38, 47] and bounding boxes [12] to achieve HOI prediction. Recent Transformer-based one-stage methods [33] have significantly improved the performance of the model by adaptively aggregating global contexts through self-attention and cross-attention mechanisms [3, 35]. However, limited by the requirements of object detection and interaction classification simultaneously, it is difficult for the one-stage paradigm to achieve a good trade-off in multi-task learning [42]. In contrast, the two-stage paradigm typically uses pre-trained object detectors (*e.g.*, Faster R-CNN [31] and DETR [3]) to form candidate human-object pairs, which are then passed to an interac-

tion classifier. Despite the performance improvement, these methods only extract visual features with predefined labels, which makes them lack the generalization ability to extend to novel interactions [24, 37].

2.2. VLM-based HOI detection

Recently, the development of large Vision-Language Models (VLMs) has shown their transfer and application ability in downstream tasks [20, 30]. In view of this, many methods try to explore the knowledge within VLMs to construct generalizable zero-shot HOI detectors. GEN-VLKT [22] initializes the interaction classifier weights through the text embedding of CLIP. EoID [39] employs a knowledge distillation strategy to transfer the distribution of action probabilities within CLIP to the HOI model. EZ-HOI [15] utilizes LLM [1, 2, 7, 11] to provide nuanced differentiation between unseen classes and their related seen classes. However, these methods still cannot escape the need for pre-trained object detectors, while assuming that unseen HOI class names are known (within the predefined list), which makes it difficult for them to handle unseen interactions outside the predefined list and limits their generalization ability in open-world scenarios [18, 23, 37].

2.3. CLIP-based Open Vocabulary HOI detection

Unlike zero-shot HOI detection, CLIP-based Open Vocabulary HOI detection prohibits the use of names of unseen classes during training and does not require pre-trained detectors. Therefore, it is more practical for open scenarios and has gradually become a relatively popular research direction [17, 18, 23, 37]. THID [37] first introduces the concept of OV-HOI detection, showing impressive recognition ability on the open-vocabulary SWIG-HOI dataset [36] by designing a HOI sequence parser combined with CLIP [30] to simultaneously detect multiple interactions. CMD-SE [17] further enriches the interaction representations by conditional multi-level decoding and the human body part descriptions generated by LLMs. INP-CC [18] proposes a method of interaction-aware prompts and concept calibration to enhance interaction understanding. SGC-Net [23] designs a multi-granularity visual feature extraction and LLM-based hierarchical text description comparison network, achieving the best performance. Despite the progress, these methods ignore the inherent problem of overfitting seen classes within the one-stage paradigm [16, 25], which restricts the improvement of the model's performance.

3. Method

In this section, we present our Semantic-Diversified and Interaction-Focused framework **(SD-IF)** for open-vocabulary HOI detection. SD-IF gets rid of the need for pre-trained object detectors and can generalize well to novel interaction categories in real-world scenarios.

3.1. Preliminaries

Baseline Revisited. Following CMD-SE [17], we first construct a CLIP-based one-stage OV-HOI detector to effectively transfer the knowledge conducive to interaction recognition. Similar to DETR [3], we treat HOI detection as an end-to-end set matching problem, thus avoiding the need for handcrafted components such as anchor generation. Specifically, given an input image $I \in \mathbb{R}^{H\times W\times 3}$, We first employ the pre-trained CLIP [30] Image encoder $E_v(\cdot)$ to extract the global visual representation F_V:

$$F_V = E_v(I), \quad F_V \in \mathbb{R}^{HW\times d_v}, \tag{1}$$

where d_v denotes the feature dimension of F_V.

We then initialize N_q learnable query embeddings $Q = \{q_i\}_{i=1}^{N_q}$, which serve as human-object prototypes. Through a stack of L transformer decoder layers D_V, the queries Q interact with global visual feature F_V to produce contextualized HOI representations $H \in \mathbb{R}^{N_q\times d_v}$:

$$H = D_V(Q, F_V). \tag{2}$$

where Q is used as a query, and F_V is used as key and value in the cross-attention mechanism of the HOI decoder D_V. Then, we feed H to two different head networks: 1) a bounding box regressor P_{det}, which predicts a confidence score c and the bounding box of the interacting human and object (b_h, b_o). 2) a linear projection P_{inter} which maps HOI representations H to the joint vision-and-text space, getting $\tilde{H} \in \mathbb{R}^{N_q\times d_t}$, where d_t denotes the dimension of text embeddings within CLIP. Similar to [17, 23, 37], we calculate its similarity with the textual semantic features $T_{hoi} \in \mathbb{R}^{N_q\times d_t}$ from the CLIP text encoder E_T for the final interaction recognition.

Overview Architecture. The overall architecture of our SD-IF is illustrated in Figure 2. Given an input image, we first extract the global visual representation F_V using the pre-trained CLIP image encoder E_V, and initialize the learnable query embeddings Q. Section 3.2 introduces the proposed **Semantic Diversification (SD) Module**, which applies reinforcement-driven stochastic perturbations under a dual-level semantic exploration mechanism to diversify the query embeddings. This design enables Q to explore beyond the semantic clusters of seen categories, thereby enhancing open-set generalization while preserving visual coherence. Section 3.3 then presents the **Interaction Focusing (IF) Module**, which employs an actor–critic reinforcement framework to adaptively refine spatial attention by linking detection features with interaction representations. This cooperative optimization facilitates the model to focus on interaction-critical regions and capture fine-grained HOI cues. Finally, Section 3.4 describes the overall training and inference procedure of SD-IF.

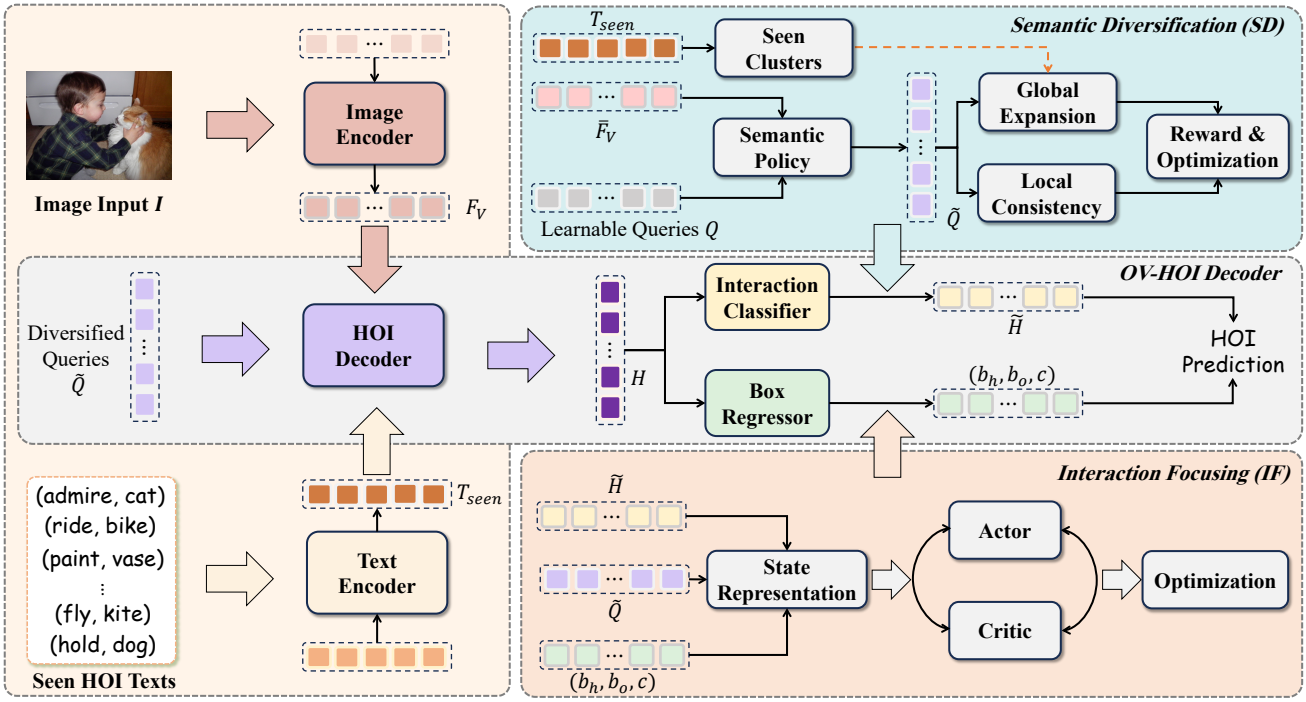


Figure 2. The framework of our SD-IF. It eliminates the need for a pretrained object detector and introduces two key modules for OV-HOI detection: (1) The Semantic Diversification (SD) module applies reinforcement-driven semantic perturbations to diversify query embeddings, enhancing open-set generalization. (2) The Interaction Focusing (IF) module adopts an actor–critic reinforcement strategy to refine spatial attention, guiding the model to focus on interaction-critical regions for more discriminative HOI reasoning.

3.2. Semantic Diversification Module (SD)

The one-stage paradigm tends to overfit to seen HOI concepts, as its query embeddings Q are optimized primarily toward interaction categories in the training set. Such concentration restricts the generalization ability to unseen categories [16, 25]. To alleviate this limitation, a novel Semantic Diversification Module (SD) is designed to treat each query $q_i \in Q$ as an agent that explores the semantic space under a learned stochastic policy, aiming to discover novel yet visually grounded semantics.

Semantic Policy Formulation. Given each learnable query embedding q_i and the global visual feature F_V, we first obtain a global visual context $\bar{F}_V = \text{AvgPool}(F_V)$ to capture image-level semantics. Then the semantic policy network g_θ outputs the mean μ_i and covariance Σ_i of a Gaussian distribution for each query:

$$\begin{aligned} a_i \sim \pi_\theta(a_i|q_i,\bar{F}_V) = \mathcal{N}(\mu_i, \Sigma_i), \\ [\mu_i, \Sigma_i] = g_\theta(q_i, \bar{F}_V), \end{aligned} \tag{3}$$

where a_i denotes the sampled semantic perturbation, g_θ is implemented with three fully connected layers. The updated query is defined as:

$$\tilde{q}_i = q_i + \kappa a_i, \tag{4}$$

where κ controls the perturbation magnitude. The perturbed queries $\tilde{Q} = \{\tilde{q}_i\}_{i=1}^{N_q}$ is then fed into subsequent transformer decoder layers for HOI reasoning.

Dual-Level Semantic Exploration. As shown in Figure 3, to effectively explore diverse semantics, SD employs a dual-level mechanism: *(1) Local Consistency.* The policy network g_θ learns perturbations conditioned on $\bar{F}_V$, ensuring that perturbed queries $\tilde{q}_i$ remain visually aligned and do not deviate from the global visual context. *(2) Global Expansion.* We encourage $\tilde{q}_i$ to move away from the centers of seen semantic clusters $\{\mu_c^{seen}\}_{c=1}^{C_s}$ derived from the CLIP text embeddings T_{seen} of training HOI categories:

$$\mathcal{L}_{\text{global}} = -\frac{1}{N_q}\sum_i \log\left(1 - \text{sim}(\tilde{q}_i, \mu_c^{seen})\right), \tag{5}$$

where $\text{sim}(\cdot,\cdot)$ is cosine similarity. This encourages queries to explore the semantic periphery of seen categories.

Proxy Semantic Reward. Since unseen categories are unavailable during training, we design a proxy semantic reward to approximate open-vocabulary generalization ability:

$$r_i = \underbrace{\text{KL}(p_{\text{text}}(q_i) \,\|\, p_{\text{text}}(\tilde{q}_i))}_{\text{semantic shift}} - \gamma \underbrace{\text{KL}(p_{\text{visual}}(\tilde{q}_i) \,\|\, p_{\text{visual}}(q_i))}_{\text{feature consistency}}, \tag{6}$$

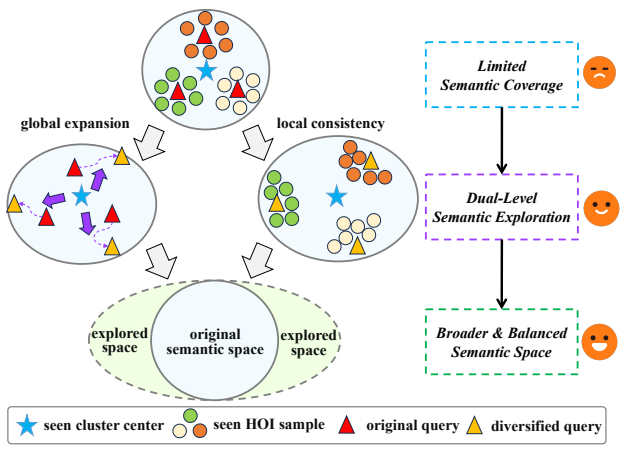


Figure 3. Illustration of the Semantic Diversification (SD) Module. The original query embeddings tend to cluster around seen HOI concepts, leading to limited semantic coverage. Through dual-level semantic exploration (comprising global expansion and local consistency), SD encourages queries to move away from seen semantic clusters while maintaining visual coherence, resulting in a broader and more balanced semantic space.

where p_{text} and p_{visual} are distributions of the query embeddings projected into the text and visual feature spaces, respectively, and γ balances the two terms. The first term encourages $\tilde{q}_i$ to deviate from seen textual semantics, while the second constrains perturbations to remain consistent with the corresponding visual domain.

Optimization. We train the policy network g_θ by maximizing the expected cumulative reward under the policy gradient objective:

$$\mathcal{L}_{local} = -\mathbb{E}_{a_i \sim \pi_\theta}[(r_i - \hat{r}_i) \log \pi_\theta(a_i|q_i)], \tag{7}$$

where $\hat{r}_i$ denotes a moving average of historical rewards to stabilize updates. The final training objective of SD combines reinforcement optimization and global regularization:

$$\mathcal{L}_{sd} = \mathcal{L}_{local} + \beta \mathcal{L}_{global}, \tag{8}$$

where β is used to control the trade-off between exploration and stability.

By combining local visual-consistent exploration and global semantic expansion, SD continuously enriches the semantic diversity of query embeddings, forming a broader and more balanced semantic space that supports open-vocabulary generalization in OV-HOI detection.

3.3. Interaction Focusing Module (IF)

While the Semantic Diversification (SD) module broadens the query distribution to enhance *semantic generalization*, the Interaction Focusing (IF) module complements it by improving *spatial discrimination*. Specifically, IF learns to attend to regions that are most indicative of interactions, mitigating the tendency of global representations to overlook fine-grained spatial cues. To this end, we formulate the focusing process as an *actor–critic reinforcement* framework that learns an adaptive attention policy that links detection cues with interaction semantics.

State Representation. Given the interaction-aligned features $\tilde{H} = \{\tilde{h}_i\}_{i=1}^{N_q}$ from the projection head P_{inter}, we construct the state representation s_i by combining diversified query semantics, detection contexts, and interaction representations:

$$s_i = \phi_s(\tilde{q}_i, p_i^{det}, \tilde{h}_i), \tag{9}$$

where $\tilde{q}_i \in \tilde{Q}$ denotes the semantically diversified query from Eq.(4), and $p_i^{det} = \{\hat{b_h^i}, \hat{b_o^i}, \hat{c}_i\}$ denotes the detection predictions from the box regressor P_{det}. This integration provides the model with a semantically grounded and spatially aware representation, bridging the gap between visual grounding and interaction reasoning.

Action Policy. Based on s_i, the actor network π_ϕ is designed to output an adaptive attention policy that highlights the interaction-critical regions:

$$A_i = \pi_\phi(s_i) = \text{Softmax}\big(f_\phi(s_i, \tilde{H})\big), \tag{10}$$

where $A_i \in \mathbb{R}^{N_q}$ denotes learned attention distribution over interaction embeddings. The f_ϕ is implemented as a multi-head cross-attention block followed by a lightweight feed-forward network and layer normalization, which takes s_i as the query, and $\tilde{H}$ as key and value. The refined interaction representation is computed as:

$$\hat{h}_i = \sum_{i=1}^{N_q} A_i \, \tilde{h}_i. \tag{11}$$

This operation allows each query to reweight semantically related interaction prototypes dynamically, thereby emphasizing interaction-critical patterns and suppressing redundant or background semantics.

Reward Design. To directly supervise the spatial grounding quality of attention, we propose a differentiable spatial focusing reward that measures the overlap between the predicted attention map A_i and the human–object region mask M_{ho} derived from detection boxes (b_h, b_o):

$$R_i^{spatial} = \frac{\sum_{x,y} A_i(x,y) \cdot M_{ho}(x,y)}{\sum_{x,y} A_i(x,y)}. \tag{12}$$

This term encourages attention to concentrate within interactive regions. To ensure semantic coherence, we further introduce a semantic consistency reward:

$$R_i^{semantic} = \text{sim}(\hat{h}_i, \tilde{q}_i), \tag{13}$$

and the final hybrid reward is defined as:

$$R_i = R_i^{spatial} + \eta R_i^{semantic}, \tag{14}$$

where η balances spatial focusing and semantic consistency. This design provides stable and interpretable reinforcement signals, guiding the actor–critic process to progressively refine the attention toward interaction-critical regions.

Optimization. The actor–critic pair (π_ϕ, C_ψ) is jointly optimized via a reinforcement learning objective:

$$\mathcal{L}_{if} = -\mathbb{E}_{A_i \sim \pi_\phi}\left[(R_i - C_\psi(s_i)) \log \pi_\phi(A_i|s_i)\right] + \|R_i - C_\psi(s_i)\|_2^2. \tag{15}$$

Through iterative actor–critic interactions, the attention policy progressively converges toward focusing on semantically coherent and spatially critical regions. By leveraging $\tilde{H}$ as the representation substrate, IF fully exploits the powerful visual-language association capabilities of CLIP and enhances generalization to unseen HOI categories.

3.4. Training and Inference

Training. The training of our network integrates supervised detection objectives $\mathcal{L}_{det}$ with reinforcement-based auxiliary optimizations from the proposed SD and IF. Specifically, we optimize our model with a composite objective:

$$\mathcal{L} = \mathcal{L}_{det} + \lambda_{sd}\mathcal{L}_{sd} + \lambda_{if}\mathcal{L}_{if}, \tag{16}$$

where λ_{sd} and λ_{if} are the hyper-parameters to adjust the weight of SD and IF reinforcement loss, respectively.

For the $\mathcal{L}_{det}$, we follow the prior query-based methods [23, 33, 42] to assign a bipartite matching prediction with each ground-truth using the Hungarian algorithm [14]. The detection matching cost $\mathcal{L}_{det}$ for the matching process and the targeting cost for the training back-propagation share the same strategy, which is formulated as follows:

$$\mathcal{L}_{det} = \lambda_b \sum_{i \in \{h,o\}} \mathcal{L}_b^i + \lambda_{iou} \sum_{i \in \{h,o\}} \mathcal{L}_{iou}^i + \lambda_{cls}\mathcal{L}_{cls}, \tag{17}$$

where $\mathcal{L}_b$, $\mathcal{L}_{iou}$, and $\mathcal{L}_{cls}$ denote the box regression, intersection over union, and classification losses, respectively. λ_b, λ_{iou}, and λ_{cls} are the hyper-parameter weight.

Inference. At inference time, the reinforcement modules act deterministically without sampling. For SD, we use the mean perturbation μ_i from Eq. (4) to obtain stabilized diversified queries:

$$\tilde{q}_i = q_i + \kappa\mu_i. \tag{18}$$

For IF, the actor directly generates the attention distribution A_i without critic feedback.

For each HOI prediction, given the bounding box score $\hat{c}_i$ from the box regressor, and the interaction score $\hat{f}_i$ from the interaction classifier, the final score $\hat{f}_i{}'$ is calculated as:

$$\hat{f}_i{}' = \hat{f}_i \cdot \hat{c}_i{}^\eta, \tag{19}$$

where $\eta > 1$ is a constant used during inference to suppress overconfident detection results [43, 44].

Method	Pretrained Detector	Unseen	Seen	Full
Zero-shot Methods				
FCL [9] (CVPR'21)	✓	13.16	24.23	22.01
SCL [10] (ECCV'22)	✓	19.07	30.39	28.08
GEN-VLKT [22] (CVPR'22)	✓	21.36	32.91	30.56
OpenCat [46] (CVPR'23)	✓	21.46	33.86	31.38
HOICLIP [28] (CVPR'23)	✓	**23.48**	**34.47**	**32.26**
Open-vocabulary Methods				
THID [37] (CVPR'22)	✗	15.53	24.32	22.38
CMD-SE [17] (CVPR'24)	✗	16.70	23.95	22.35
INP-CC [18] (ICCV'25)	✗	17.38	24.74	23.12
SGC-Net [23] (CVPR'25)	✗	23.27	28.34	27.22
SD-IF (Ours)	✗	**28.01**	**30.09**	**29.63**

Table 1. Comparison of our proposed SD-IF with state-of-the-art methods on the HICO-DET dataset under the simulated open vocabulary setting.

4. Experiment

4.1. Experimental Setting

Datasets. We evaluate the proposed SD-IF on two public benchmarks, HICO-DET [4] and SWIG-HOI [36]. The HICO-DET labels 80 object classes, 117 interaction classes, and 600 HOI categories. Following previous works[17, 18, 23, 37], we hold out 120 rare HOIs to simulate the open vocabulary setting. The SWIG-HOI has unseen HOI classes naturally in the test set. At inference time, SWIG-HOI requires a comprehensive evaluation of performance on 1391 unseen, 2841 rare, and 1307 non-rare HOI classes, which puts a high demand on the generalization of the model.

Evaluation Metric. We use the mean Average Precision (mAP) as the evaluation metric on these datasets. Following prior works [13, 17, 28, 37, 44], a predicted HOI triplet is regarded as a true-positive HOI sample when the Intersection over Union (IoU) between the detected human and object bounding boxes and the corresponding ground-truth boxes is larger than 0.5, and the interaction class prediction is correct simultaneously.

Implementation Details. Following [17, 37], we employ the CLIP ViT-B/16 version as our visual encoder, and introduce 8 prefix tokens and 4 conjunctive tokens to connect the words of actions and objects within the text encoder. The SD-IF is optimized using AdamW with an initial learning rate of 10^{-4}. We set the hyperparameter λ_{sd} and λ_{if} in Eq. (16) to 1 and 2, respectively. Following [17, 23], we set the detection cost weights to $\mathcal{L}_b$, $\mathcal{L}_{iou}$, and $\mathcal{L}_{cls}$ to 5, 2, and 5, respectively. The hyperparameters κ, γ, and β of the SD module are set to 0.2, 0.5, and 0.5, respectively. We set the hyperparameter η in Eq.(14) as 0.2. Our SD-IF is trained for 80 epochs with a batch size of 128 on 2 A100 GPUs.

4.2. Comparisons with State-of-the-Art Methods

HICO-DET. As shown in Table 1, we compare the performance of the proposed SD-IF with the previous SOTA

Method	Non-rare	Rare	Unseen	Full
CHOID [36] (ICCV'21)	10.93	6.63	2.64	6.64
QPIC [33] (CVPR'21)	16.95	10.84	6.21	11.12
GEN-VLKT [22] (CVPR'22)	20.91	10.41	-	10.87
MP-HOI [41] (CVPR'24)	20.28	14.78	-	12.61
THID [37] (CVPR'22)	17.67	12.82	10.04	13.26
CMD-SE [17] (CVPR'24)	21.46	14.64	10.70	15.26
INP-CC [18] (ICCV'25)	22.84	16.74	11.02	16.74
SGC-Net [23] (CVPR'25)	23.67	16.55	12.46	17.20
SD-IF (Ours)	**25.01**	**18.86**	**15.57**	**19.48**

Table 2. Comparison of our proposed SD-IF with state-of-the-art methods on the SWIG-HOI dataset.

Method	Non-rare	Rare	Unseen	Full
Base	15.75	11.51	7.35	11.47
+ *SD*	21.39	16.23	14.01	16.89
+ *IF*	24.55	16.49	13.50	17.64
Ours	**25.01**	**18.86**	**15.57**	**19.48**

Table 3. Ablation studies of different modules of our SD-IF on the SWIG-HOI dataset. SD denotes the Semantic Diversification Module, IF denotes the Interaction Focusing Module.

zero-shot methods as well as the OV-HOI methods in the simulated open-vocabulary setting of the HICO-DET. It can be seen that our SD-IF outperforms SGC-Net [23] by 4.74%, 1.75%, and 2.41% in mAP on the unseen, seen, and full categories, respectively. This demonstrates the effective interaction awareness and strong generalization ability of our method. Notably, as shown in Figure 1 (a), our approach alleviates the performance gap between the seen and unseen classes, narrowing the difference between them from 5.07% to 2.08%, mainly thanks to the introduction of the Semantic Diversification (SD) module, which expands the semantic coverage of the queries and weakens their overfitting to the seen interactions. Furthermore, the zero-shot methods outperform the open-vocabulary methods in overall performance due to the presence of a pre-trained detector. However, the proposed SD-IF still surpasses these methods in the performance of unseen classes even without a pre-trained object detector, further demonstrating the superiority of our method.

SWIG-HOI. We further conduct a comprehensive evaluation of the generalization ability of the network in SWIG-HOI. As shown in Table 2, compared with the best OV-HOI detection method SGC-Net [23], SD-IF achieves 1.34%, 2.31%, 3.11%, and 2.28% performance improvement on non-rare, rare, unseen, and full categories, respectively. The improvement is most significant on rare (relative 14.0%) and unseen (relative 25.0%) categories, demonstrating the effectiveness of our method in improving the generalization. In addition, existing methods such as GEN-VLKT [22] and MP-HOI [41], which rely on pre-trained object detectors, perform poorly on the SWIG-HOI dataset because they cannot effectively expand the interaction categories. In contrast, SD-IF completely discards the need for a pre-trained detector and outperforms these methods in all open-vocabulary metrics, further demonstrating its great potential for applications in open scenarios.

λ_{sd}	λ_{if}	Non-rare	Rare	Unseen	Full
0.0	0.0	15.75	11.51	7.35	11.47
1.0	1.0	23.57	16.53	13.99	17.55
1.0	2.0	**25.01**	**18.86**	**15.57**	**19.48**
1.0	3.0	24.58	17.79	15.06	18.71
2.0	2.0	24.77	17.40	15.07	18.55

Table 4. Ablations studies of the hyperparemeters of Eq.(16) on the SWIG-HOI dataset.

LC	GE	Non-rare	Rare	Unseen	Full
		24.55	16.49	13.50	17.64
✓		24.78	17.01	13.79	18.03
	✓	24.84	17.82	15.43	18.88
✓	✓	**25.01**	**18.86**	**15.57**	**19.48**

Table 5. Comparison of different semantic exploration strategies of SD module on the SWIG-HOI dataset. LC: Local Consistency. GE: Global Expansion.

SFR	SCR	Non-rare	Rare	Unseen	Full
		21.39	16.23	14.01	16.89
✓		24.83	18.70	15.39	19.32
	✓	22.66	16.82	14.73	17.67
✓	✓	**25.01**	**18.86**	**15.57**	**19.48**

Table 6. Comparison of different rewards of IF module on the SWIG-HOI dataset. SFR: Spatial Focusing Reward. SCR: Semantic Consistency Reward.

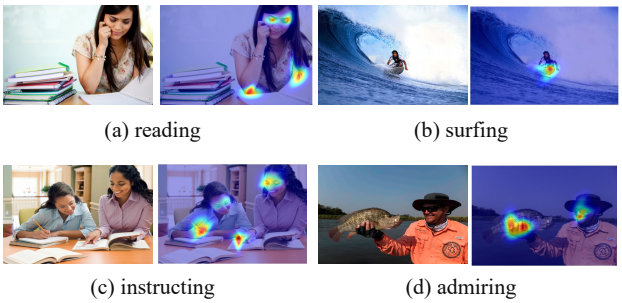
Method	Non-rare	Rare	Unseen	Full
Base+SD	21.39	16.23	14.01	16.89
+ *IF (PPO)*	23.58	17.59	14.88	18.32
+ *IF (SAC)*	24.57	17.43	14.92	18.48
+ *IF (our AC)*	**25.01**	**18.86**	**15.57**	**19.48**

Table 7. Ablation studies with advanced RL algorithms.

4.3. Ablation Study

To analyze the rationality and effectiveness of different components or parameters within the proposed SD-IF, we conduct ablation experiments on the SWIG-HOI dataset, with results shown in Table 3-7.

Overall Model Framework Design. As shown in Table 3, we first analyze the contribution of the proposed Semantic Diversification (SD) and Interaction Focusing (IF) modules to the overall performance. Our baseline model follows the configuration of CMD-SE [17], which lever-



(a) reading (b) surfing

(c) instructing (d) admiring

Figure 4. Qualitative results of our SD-IF on SWIG-HOI test set. SD-IF has powerful interaction perception and recognition ability in complex and variable scenes, which focuses on the interaction-relevant regions and mine fine-grained interaction cues.

ages CLIP for feature extraction and employs a transformer decoder to predict HOI triplets, thereby eliminating the need for pre-trained object detectors. By introducing the SD module (+*SD*), the model exhibits notable gains of 5.64%, 4.72%, 6.66%, and 5.42% in non-rare, rare, unseen, and full categories, respectively, demonstrating its effectiveness in enhancing generalization to novel HOI categories. When incorporating the IF module (+*IF*) into the baseline, the model achieves a 17.64% mAP on the full set, highlighting the benefit of reinforcing spatial discrimination through adaptive attention refinement. Finally, combining both modules yields the best overall performance, as the SD module expands the model's semantic perception boundary at the semantic level, while the IF module strengthens the capture of fine-grained interaction cues at the spatial level, together showing strong potential for robust HOI understanding in complex real-world scenes.

Loss hyperparameters λ_{sd} and λ_{if}. As shown in Table 4, $\lambda_{sd}, \lambda_{if}$ set to 1.0 and 2.0 yield the best results in Eq. (16), indicating that a slightly higher weight on the IF loss leads to more discriminative interaction features.

Comparison of different semantic exploration in SD. To enrich query semantics, the SD module employs dual-level exploration: local consistency (*LC*) preserves visual alignment, while global expansion (*GE*) pushes queries away from seen semantic clusters. As shown in Table 5, *LC* mainly benefits non-rare classes, whereas *GE* brings a significant +1.93 mAP gain on unseen classes. Combining both yields the best overall performance, confirming their complementary effects.

Effect of different reward designs in IF. In the IF module, we introduce a spatial focusing reward (*SFR*, Eq. 12) and a semantic consistency reward (*SCR*, Eq. 13) to guide fine-grained interaction learning. As Table 6 shows, *SFR* brings significant gains across metrics, while *SCR* offers moderate improvements. Combining both yields the best performance, indicating their complementary roles in enhancing interaction perception.

Ablation with advanced RL algorithms (PPO and SAC). To verify that the performance gain of IF does not come merely from using reinforcement learning, we replace our lightweight actor–critic policy with two widely adopted RL algorithms: Proximal Policy Optimization (PPO) [32] and Soft Actor-Critic (SAC) [8]. As shown in Table 7, both PPO and SAC underperform our formulation across all open-vocabulary metrics. PPO slightly improves over the baseline but remains inferior to our method due to its clipped objective, which restricts policy updates and limits its ability to refine fine-grained interaction regions. SAC, although effective for continuous control, introduces excessive entropy-driven exploration that causes the attention policy to drift toward irrelevant regions, reducing spatial discrimination. In contrast, our task-specific actor–critic design provides stable and visually grounded policy updates, enabling the attention to precisely focus on interaction-critical areas. These results confirm that our tailored RL formulation is better aligned with the structured spatial–semantic nature of HOI.

4.4. Qualitative Results

Our model exhibits strong open-vocabulary interaction recognition and perception capabilities. As shown in Figure 4 (a), for the *reading* action, the model precisely attends to the eye region, effectively capturing the key interaction cues required for this behavior. In Figure 4 (b), under a small-object scenario, it accurately emphasizes the contact area between the human and the object, demonstrating sensitivity to fine-grained spatial relationships. In Figure 4 (c), which involves human–human interactions, the model correctly focuses on the eye and hand regions associated with the *instructing* action, reflecting its strong capability for understanding complex interpersonal interactions. Moreover, for the *admiring* action in Figure 4 (d), our method highlights the contact points, eyes, and smiling mouth region — all of which are crucial semantic cues for accurately recognizing *admiring*. Overall, SD-IF effectively captures discriminative and fine-grained interaction cues across diverse and complex scenes, exhibiting superior spatial focusing and semantic comprehension abilities.

5. Conclusion

In this paper, we present SD-IF, a novel framework designed to address two critical challenges in open-vocabulary HOI detection: semantic overfitting to seen categories and insufficient spatial discrimination. Specifically, the Semantic Diversification (SD) module introduces reinforcement-driven stochastic perturbations and dual-level semantic exploration to expand the query embedding manifold beyond seen semantic clusters while maintaining visual coherence. Complementarily, the Interaction Focusing (IF) module employs an actor–critic reinforcement mechanism to adaptively refine attention toward interaction-critical regions, effectively

capturing fine-grained spatial cues. Through extensive experiments and ablation studies on two public benchmarks, we demonstrate that SD-IF achieves state-of-the-art performance, validating the effectiveness of integrating semantic diversification and spatial focusing for robust open-vocabulary HOI understanding.

References

[1] Stella Biderman, Hailey Schoelkopf, Quentin Gregory Anthony, Herbie Bradley, Kyle O'Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, USVSN Sai Prashanth, Edward Raff, et al. Pythia: A suite for analyzing large language models across training and scaling. In *International Conference on Machine Learning*, pages 2397–2430. PMLR, 2023. 3

[2] Tom B Brown. Language models are few-shot learners. *arXiv preprint arXiv:2005.14165*, 2020. 3

[3] Nicolas Carion, Francisco Massa, Gabriel Synnaeve, Nicolas Usunier, Alexander Kirillov, and Sergey Zagoruyko. End-to-end object detection with transformers. In *European conference on computer vision*, pages 213–229. Springer, 2020. 2, 3

[4] Yu-Wei Chao, Yunfan Liu, Xieyang Liu, Huayi Zeng, and Jia Deng. Learning to detect human-object interactions. In *2018 ieee winter conference on applications of computer vision (wacv)*, pages 381–389. IEEE, 2018. 2, 6

[5] Po-Yi Chen, Alexander H Liu, Yen-Cheng Liu, and Yu-Chiang Frank Wang. Towards scene understanding: Unsupervised monocular depth estimation with semantic-aware representation. In *Proceedings of the IEEE/CVF Conference on computer vision and pattern recognition*, pages 2624–2632, 2019. 1

[6] Li Dong, Nan Yang, Wenhui Wang, Furu Wei, Xiaodong Liu, Yu Wang, Jianfeng Gao, Ming Zhou, and Hsiao-Wuen Hon. Unified language model pre-training for natural language understanding and generation. *Advances in Neural Information Processing Systems*, 32, 2019. 2

[7] Danny Driess, Fei Xia, Mehdi SM Sajjadi, Corey Lynch, Aakanksha Chowdhery, Brian Ichter, Ayzaan Wahid, Jonathan Tompson, Quan Vuong, Tianhe Yu, et al. Palm-e: An embodied multimodal language model. *arXiv preprint arXiv:2303.03378*, 2023. 3

[8] Tuomas Haarnoja, Aurick Zhou, Pieter Abbeel, and Sergey Levine. Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor. In *International conference on machine learning*, pages 1861–1870. Pmlr, 2018. 8

[9] Zhi Hou, Baosheng Yu, Yu Qiao, Xiaojiang Peng, and Dacheng Tao. Detecting human-object interaction via fabricated compositional learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 14646–14655, 2021. 6

[10] Zhi Hou, Baosheng Yu, and Dacheng Tao. Discovering human-object interaction concepts via self-compositional learning. In *European Conference on Computer Vision*, pages 461–478. Springer, 2022. 6

[11] Srinivasan Iyer, Xi Victoria Lin, Ramakanth Pasunuru, Todor Mihaylov, Daniel Simig, Ping Yu, Kurt Shuster, Tianlu Wang, Qing Liu, Punit Singh Koura, et al. Opt-iml: Scaling language model instruction meta learning through the lens of generalization. *arXiv preprint arXiv:2212.12017*, 2022. 3

[12] Bumsoo Kim, Taeho Choi, Jaewoo Kang, and Hyunwoo J Kim. Uniondet: Union-level detector towards real-time human-object interaction detection. In *Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XV 16*, pages 498–514. Springer, 2020. 2

[13] Bumsoo Kim, Jonghwan Mun, Kyoung-Woon On, Minchul Shin, Junhyun Lee, and Eun-Sol Kim. Mstr: Multi-scale transformer for end-to-end human-object interaction detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 19578–19587, 2022. 6

[14] Harold W Kuhn. The hungarian method for the assignment problem. *Naval research logistics quarterly*, 2(1-2):83–97, 1955. 6

[15] Qinqian Lei, Bo Wang, and Robby Tan. Ez-hoi: Vlm adaptation via guided prompt learning for zero-shot hoi detection. *Advances in Neural Information Processing Systems*, 37:55831–55857, 2025. 3

[16] Ting Lei. Exploring conditional multi-modal prompts for zero-shot hoi detection. In *ECCV*, pages 1–19, 2024. 2, 3, 4

[17] Ting Lei, Shaofeng Yin, and Yang Liu. Exploring the potential of large foundation models for open-vocabulary hoi detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 16657–16667, 2024. 2, 3, 6, 7

[18] Ting Lei, Shaofeng Yin, Qingchao Chen, Yuxin Peng, and Yang Liu. Open-vocabulary hoi detection with interaction-aware prompt and concept calibration. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 23945–23957, 2025. 2, 3, 6, 7

[19] Juncheng Li, Xin He, Longhui Wei, Long Qian, Linchao Zhu, Lingxi Xie, Yueting Zhuang, Qi Tian, and Siliang Tang. Fine-grained semantically aligned vision-language pre-training. *Advances in Neural Information Processing Systems*, 35:7290–7303, 2022. 2

[20] Junnan Li, Dongxu Li, Caiming Xiong, and Steven Hoi. Blip: Bootstrapping language-image pre-training for unified vision-language understanding and generation. In *International conference on machine learning*, pages 12888–12900. PMLR, 2022. 2, 3

[21] Yue Liao, Si Liu, Fei Wang, Yanjie Chen, Chen Qian, and Jiashi Feng. Ppdm: Parallel point detection and matching for real-time human-object interaction detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 482–490, 2020. 1, 2

[22] Yue Liao, Aixi Zhang, Miao Lu, Yongliang Wang, Xiaobo Li, and Si Liu. Gen-vlkt: Simplify association and enhance interaction understanding for hoi detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 20123–20132, 2022. 3, 6, 7

[23] Xin Lin, Chong Shi, Zuopeng Yang, Haojin Tang, and Zhili Zhou. Sgc-net: Stratified granular comparison network

for open-vocabulary hoi detection. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 4539–4549, 2025. 2, 3, 6, 7

[24] Jinguo Luo, Weihong Ren, Weibo Jiang, Xi'ai Chen, Qiang Wang, Zhi Han, and Honghai Liu. Discovering syntactic interaction clues for human-object interaction detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 28212–28222, 2024. 3

[25] Yunyao Mao, Jiajun Deng, Wengang Zhou, Li Li, Yao Fang, and Houqiang Li. Clip4hoi: towards adapting clip for practical zero-shot hoi detection. *Advances in Neural Information Processing Systems*, 36:45895–45906, 2023. 2, 3, 4

[26] Esteve Valls Mascaro, Daniel Sliwowski, and Dongheui Lee. Hoi4abot: Human-object interaction anticipation for human intention reading collaborative robots. *arXiv preprint arXiv:2309.16524*, 2023. 1

[27] Jean Massardi, Mathieu Gravel, and Éric Beaudry. Parc: A plan and activity recognition component for assistive robots. In *2020 IEEE International Conference on Robotics and Automation (ICRA)*, pages 3025–3031. IEEE, 2020. 1

[28] Shan Ning, Longtian Qiu, Yongfei Liu, and Xuming He. Hoiclip: Efficient knowledge transfer for hoi detection with vision-language models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 23507–23517, 2023. 6

[29] Songyou Peng, Kyle Genova, Chiyu Jiang, Andrea Tagliasacchi, Marc Pollefeys, Thomas Funkhouser, et al. Openscene: 3d scene understanding with open vocabularies. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 815–824, 2023. 1

[30] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International conference on machine learning*, pages 8748–8763. PMLR, 2021. 2, 3

[31] Shaoqing Ren, Kaiming He, Ross Girshick, and Jian Sun. Faster r-cnn: Towards real-time object detection with region proposal networks. *IEEE transactions on pattern analysis and machine intelligence*, 39(6):1137–1149, 2016. 2

[32] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017. 8

[33] Masato Tamura, Hiroki Ohashi, and Tomoaki Yoshinaga. Qpic: Query-based pairwise human-object interaction detection with image-wide contextual information. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10410–10419, 2021. 2, 6, 7

[34] Oytun Ulutan, ASM Iftekhar, and Bangalore S Manjunath. Vsgnet: Spatial attention network for detecting human object interactions using graph convolutions. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 13617–13626, 2020. 1

[35] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017. 2

[36] Suchen Wang, Kim-Hui Yap, Henghui Ding, Jiyan Wu, Junsong Yuan, and Yap-Peng Tan. Discovering human interactions with large-vocabulary objects via query and multi-scale detection. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 13475–13484, 2021. 2, 3, 6, 7

[37] Suchen Wang, Yueqi Duan, Henghui Ding, Yap-Peng Tan, Kim-Hui Yap, and Junsong Yuan. Learning transferable human-object interaction detector with natural language supervision. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 939–948, 2022. 2, 3, 6, 7

[38] Tiancai Wang, Tong Yang, Martin Danelljan, Fahad Shahbaz Khan, Xiangyu Zhang, and Jian Sun. Learning human-object interaction detection using interaction points. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 4116–4125, 2020. 2

[39] Mingrui Wu, Jiaxin Gu, Yunhang Shen, Mingbao Lin, Chao Chen, and Xiaoshuai Sun. End-to-end zero-shot hoi detection via vision and language knowledge distillation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 2839–2846, 2023. 3

[40] Nan Xi, Jingjing Meng, and Junsong Yuan. Open set video hoi detection from action-centric chain-of-look prompting. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 3079–3089, 2023. 1

[41] Jie Yang, Bingliang Li, Ailing Zeng, Lei Zhang, and Ruimao Zhang. Open-world human-object interaction detection via multi-modal prompts. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 16954–16964, 2024. 7

[42] Aixi Zhang, Yue Liao, Si Liu, Miao Lu, Yongliang Wang, Chen Gao, and Xiaobo Li. Mining the benefits of two-stage and one-stage hoi detection. *Advances in Neural Information Processing Systems*, 34:17209–17220, 2021. 1, 2, 6

[43] Frederic Z Zhang, Dylan Campbell, and Stephen Gould. Spatially conditioned graphs for detecting human-object interactions. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 13319–13327, 2021. 6

[44] Frederic Z Zhang, Dylan Campbell, and Stephen Gould. Efficient two-stage detection of human-object interactions with a novel unary-pairwise transformer. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 20104–20112, 2022. 1, 6

[45] Minghang Zheng, Xinhao Cai, Qingchao Chen, Yuxin Peng, and Yang Liu. Training-free video temporal grounding using large-scale pre-trained models. In *European Conference on Computer Vision*, pages 20–37. Springer, 2024. 1

[46] Sipeng Zheng, Boshen Xu, and Qin Jin. Open-category human-object interaction pre-training via language modeling framework. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 19392–19402, 2023. 6

[47] Xubin Zhong, Xian Qu, Changxing Ding, and Dacheng Tao. Glance and gaze: Inferring action-aware points for one-stage human-object interaction detection. In *Proceedings of*

the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pages 13234–13243, 2021. 2